

Physio. Q- Level 4-CARDIOTHORACIC MEDICIN and SURGERY

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1. List four factors affecting pulmonary blood pressure and flow:

1. Automatic Control {Alveolar Oxygen}:
2. Cardiac output.
- 3 .Left atrial pressure
4. Gravity
- 5 .Nervous control
- 6 . Chemical substances
7. Respiratory movements

2. What is the meaning of Ventilation/Perfusion Ratio:

It denotes the ratio of alveolar ventilation and the amount of blood that perfuse the alveoli.
Mathematically, $V/Q = 0.84$

3- Why the pulmonary artery BP is lower than the systemic arterial BP ?

- a) The output of the right ventricle is less than that of the left ventricle.
- b) The left ventricular wall is thicker than that of the right ventricle. d
- c) The aorta is more elastic than the pulmonary artery.
- d) The pulmonary resistance is less than the systemic resistance.

4- Which of the following is true regarding pulmonary circulation?

- A. It is a low-pressure system that allows for improved gas exchange.
 - B. It is the larger of the two circulatory systems.
 - C. The system functions with an increased arterial pressure to circulate through the distal parts of the body.
 - D. It consists of the left side of the heart, the aorta, and its branches.
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1- Determine the ionic basis of ventricular action potential

Phase 0: Initial rapid depolarization: due to Na^+ inflow due to opening of fast Na^+ channels.

Phase 1: Brief initial repolarization: due to closure of Na^+ channels, little K^+ efflux and Cl^- influx.

Phase 2: Prolonged plateau: due to opening of slow prolonged Ca^{++} channels.(A balance is created between influx of Ca^{++} and outflux of K^+). So, the potential is kept at 0mV

Phase 3: Late rapid repolarization: due to prolonged opening of K^+ channels and closure of Ca^{++} channels.

Phase 4: Resting membrane potential is restored (-90 mv) by $Na^+ - K^+$ pump.

2- Demonstrate Relationship Between Heart Rate (HR), Cardiac Filling, and Cardiac Output (CO) Figure 6:

Excessive tachycardia ≥ 200 beats /min. decreases CO by $\downarrow$ preload and stroke volume.

It also impairs cardiac function from decreased coronary perfusion in shortened diastole.

Sever bradycardia ≤ 50 beats/minute

decreases cardiac output. Although stroke

volume $\uparrow$ but this increase will be insufficient to balance the excessive decrease in the heart rate.

Mild decrease from 70-50 beats/minute, the ventricles have enough time for maximum filling. So, the stroke volume $\uparrow$ and the cardiac output remains constant.

3- A 75-year-old woman with a history of chronic congestive heart failure undergoes cardiac catheterization to determine her extent of cardiac dysfunction. During estimation of her systolic function, in what phase of the cardiac cycle should her left ventricular volume unchanged?

A. Rapid filling

B. Isovolumetric contraction

C. Ventricular ejection

D. Atrial systole

4- What is the cardiac ejection fraction?

A. It is the percentage ratio of cardiac work to total energy expenditure.

B. It is the cardiac output per square meter body surface area.

C. It is the percentage of blood leaving the heart in each cardiac cycle.

D. It is a premature contraction of the heart occurring during diastole.

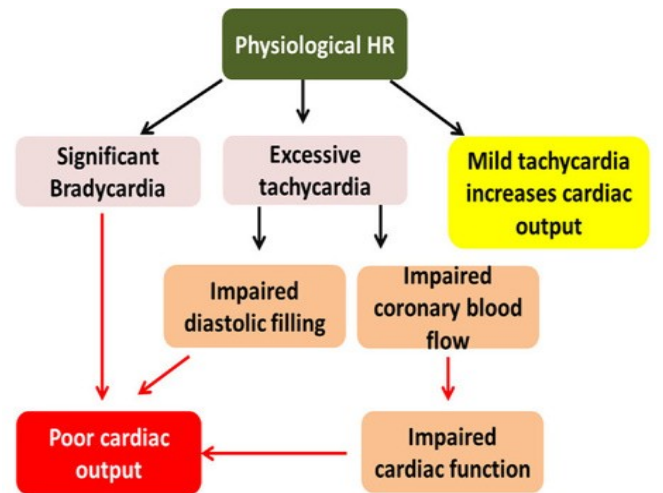


Fig.6
Relationship between heart rate (HR), cardiac filling and cardiac output (CO)